

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science focused on building systems capable of performing tasks that normally require human intelligence. Examples include image recognition, speech recognition, recommendation systems, and robotics.

Machine Learning

Machine Learning (ML) is a subset of AI in which algorithms learn patterns from data. Common learning paradigms include supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning

Deep Learning uses neural networks with many layers to solve complex tasks. It is widely used in computer vision, natural language processing, and speech recognition.

Transformers

The Transformer architecture introduced the self-attention mechanism, allowing models to process relationships between words efficiently. GPT and BERT are based on Transformers.

Retrieval-Augmented Generation (RAG)

RAG combines semantic search with a large language model. Documents are split into chunks, converted into embeddings, stored in a vector database such as FAISS, and the most relevant chunks are retrieved to help the language model generate accurate answers.